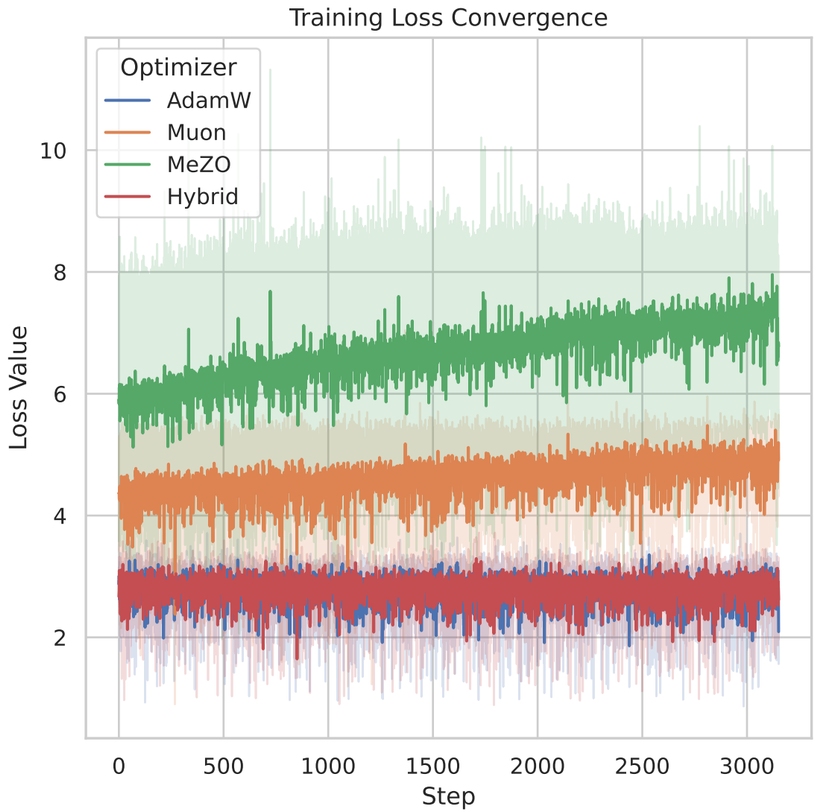
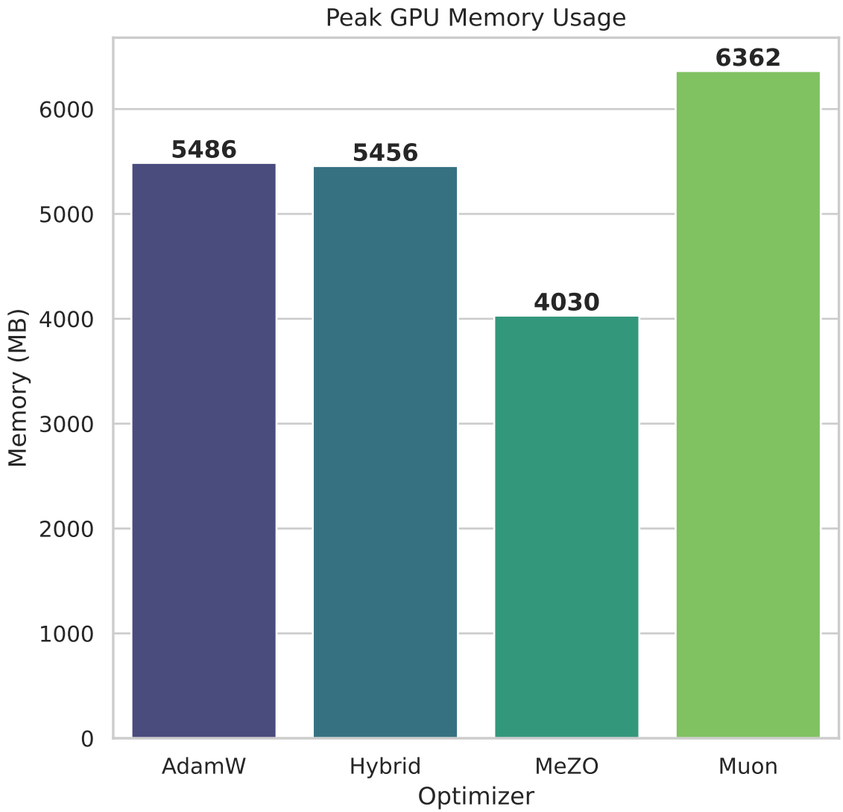
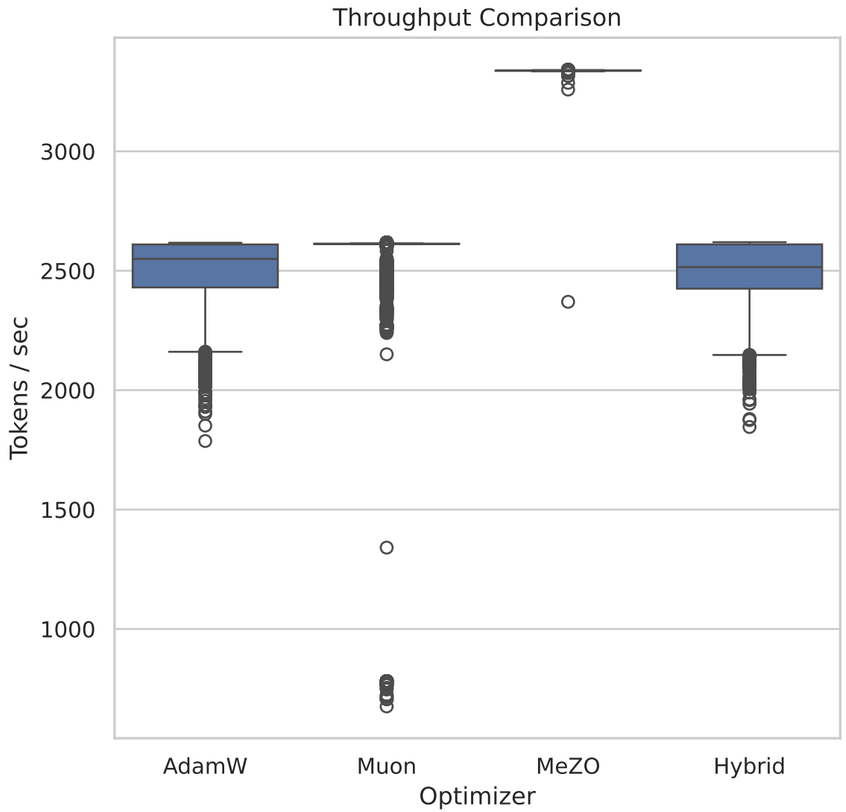

Training Loss Convergence
Optimizer
AdamW
Muon
MeZO
Hybrid
Loss Value
Step
0
500
1000
1500
2000
2500
3000
2
4
6
8
10


Peak GPU Memory Usage
Memory (MB)
Optimizer
5486
5456
4030
6362
AdamW
Hybrid
MeZO
Muon
0
1000
2000
3000
4000
5000
6000


Throughput Comparison
Tokens / sec
Optimizer
AdamW
Muon
MeZO
Hybrid
1000
1500
2000
2500
3000